



Auditing the Home Credit Default Risk ADS

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Why this ADS?

The case

Home Credit Default Risk (Kaggle). We audit *Will Koehrsen's* "Start Here: A Gentle Introduction" — the most-upvoted public solution (leaderboard **0.756**).

Why it matters

It sits at the intersection of three course themes: algorithmic fairness across demographic subgroups, the use of alternative data in high-stakes decisions, and the auditing of opaque ADS.

The core tension

Financial inclusion vs. financial risk

Too strict

Capable unbanked borrowers are denied — defeats the inclusion mission.

Too generous

Lender absorbs losses; borrowers fall into unaffordable debt.

The audit asks:

Where does this model land — and for whom?

The data, in numbers

307K

loan applications

real, operational data

122

raw features

→ 243 after encoding

92 / 8

class imbalance

repaid vs. defaulted (%)

67

columns with missing values

some > 60% missing

Critical limitation — selection bias

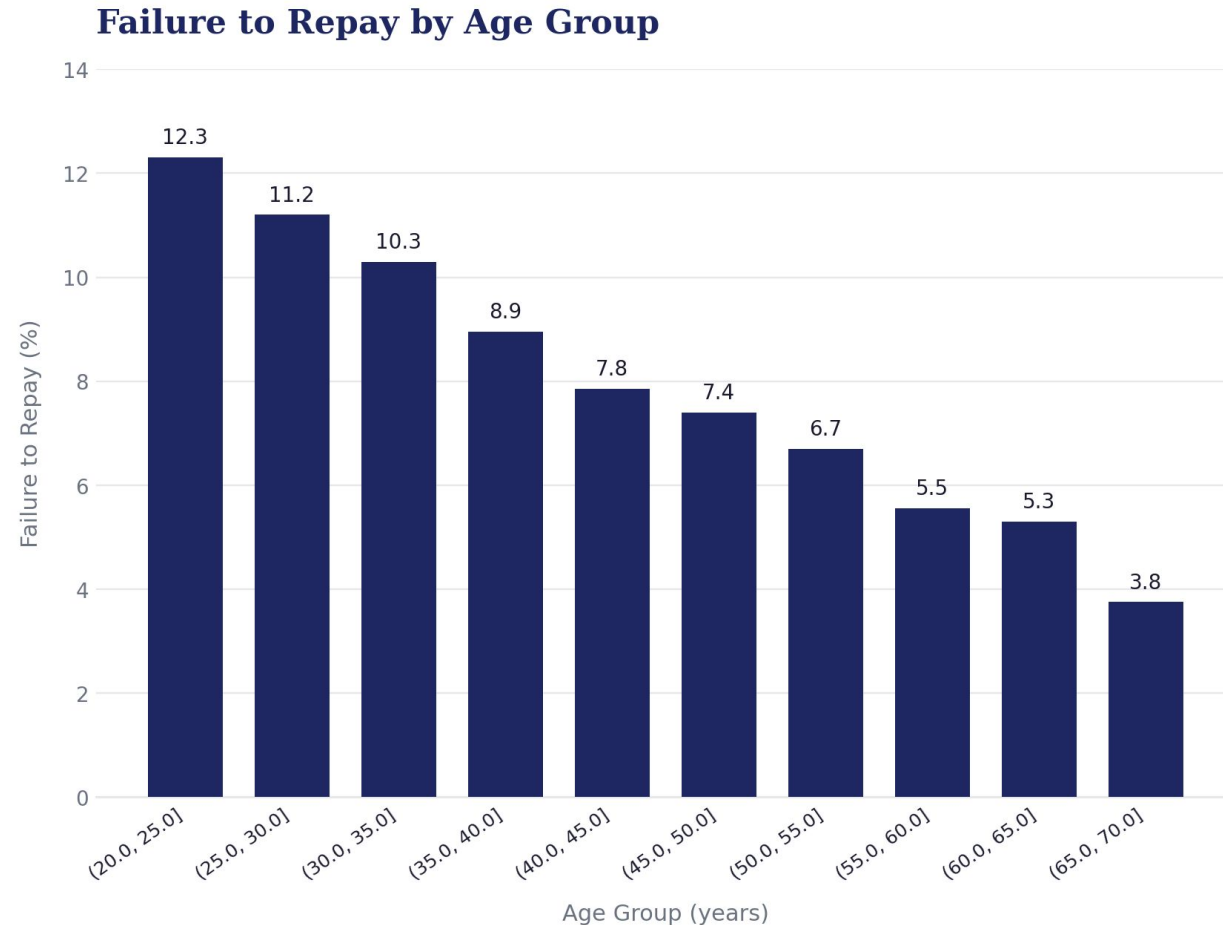
The dataset only contains people who *applied* for a Home Credit loan, and TARGET is observed only for those who were *approved*. The model learns from a survivor population shaped by past lending decisions — predictions for genuinely new unbanked populations may not generalize.

The features that drive predictions

Feature	Type	Missing	Corr. w/ TARGET	Role in the model
EXT_SOURCE_1	float [0,1]	56.4%	-0.16	External credit score (opaque)
EXT_SOURCE_2	float [0,1]	0.2%	-0.16	External credit score (opaque)
EXT_SOURCE_3	float [0,1]	19.8%	-0.18 (strongest)	External credit score (opaque)
DAYS_BIRTH	integer	0%	+0.08	Age proxy — younger = riskier
CODE_GENDER	categorical	~0%	weak	66% F · 34% M — sensitive attr.
AMT_INCOME_TOTAL	float	0%	weak	Outliers > 100M (use quartiles)

Tension: the three EXT_SOURCE variables are the strongest predictors — yet they're external, opaque, and exactly the scores *unbanked* applicants are most likely to be missing. The model leans hardest on the signal the target population lacks.

Younger applicants default more



What we see

Clear monotonic trend

Default rate falls from 12.3% (20-25) to 3.7% (65-70).

Three youngest groups

All above 10% failure rate.

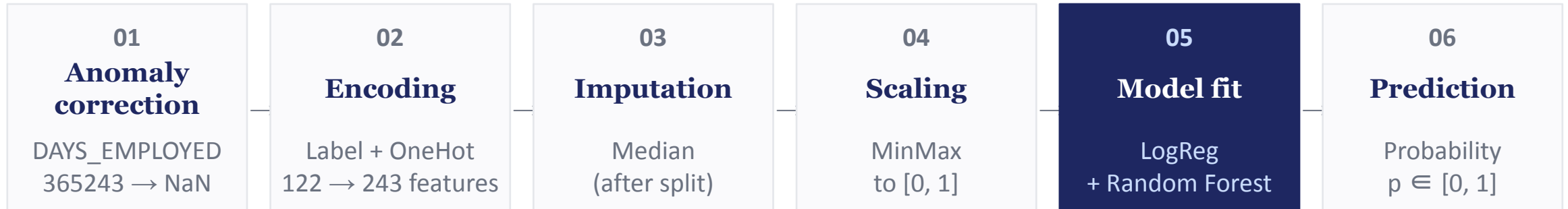
Oldest group

Below 5%.

Why it matters

Age correlates with default. Any model trained on this data risks systematic age bias — which we confirm in the audit.

How the ADS works



Two models trained

Logistic Regression $C=0.0001$, $max_iter=1000$
ROC-AUC = **0.688**

Random Forest 100 trees, unlimited depth
ROC-AUC = **0.706**

Bug we caught

SK_ID_CURR — the loan-application ID — appears in the top-10 Random Forest features with importance **0.030**.

An identifier should have zero predictive power. The model is fitting noise — the column should have been dropped before training.

Validation only goes so far

Original validation

Train on application_train.csv → submit to Kaggle → ROC-AUC.

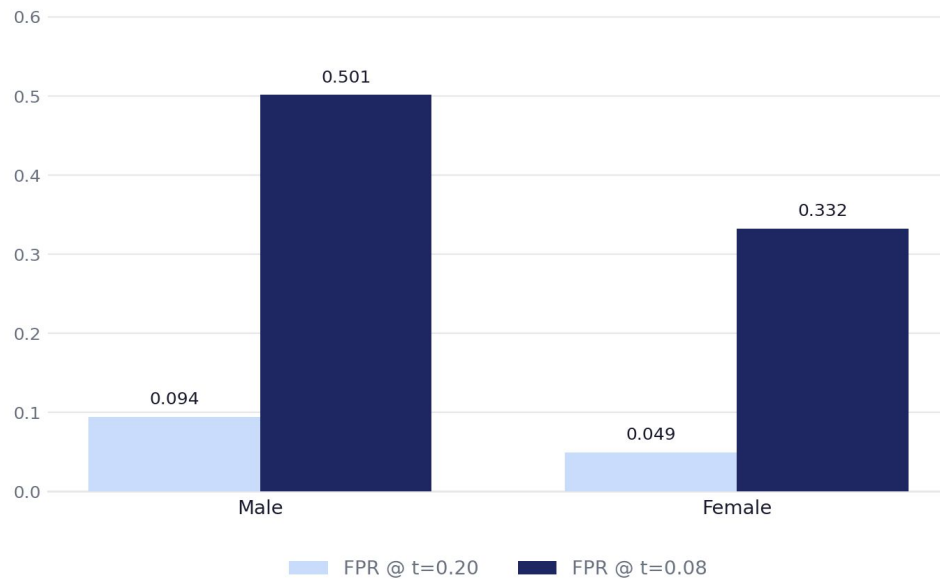
- ✓ Handles 92:8 class imbalance — accuracy alone would score 92% by always predicting "no default."
- ✓ Measures ranking quality without picking a threshold.
- ✓ Our 80/20 stratified split reproduces the Kaggle baseline (RF = 0.706).

But it does not establish...

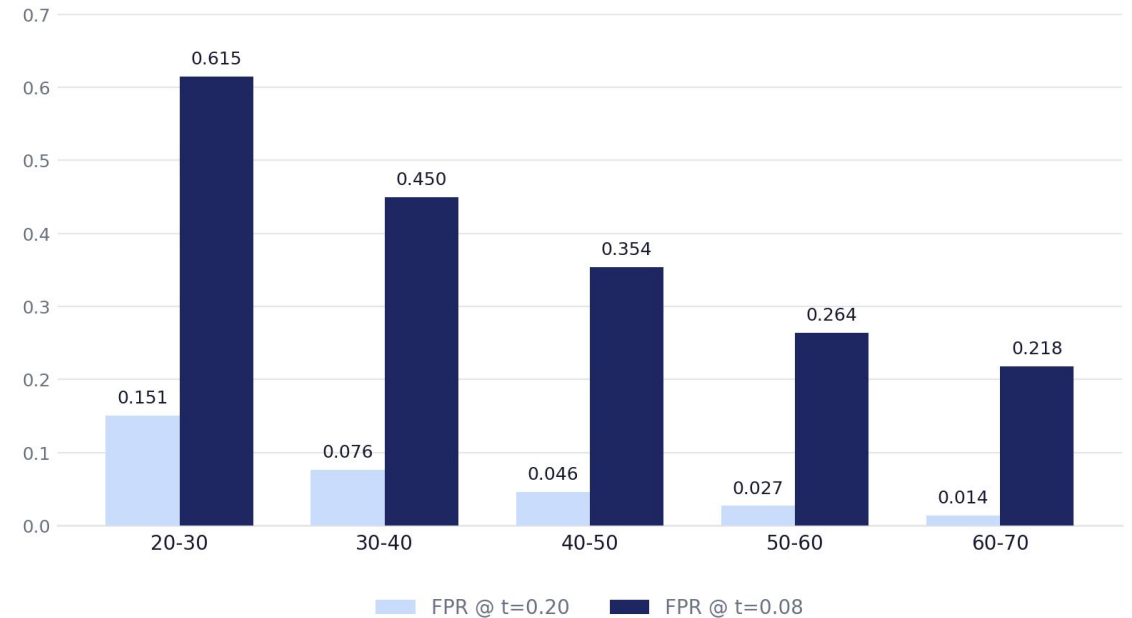
- ✗ Fairness across demographic subgroups
- ✗ Calibration — does $p = 0.3$ really mean a 30% default chance?
- ✗ Interpretability — what features actually drive a prediction?
- ✗ Reliability for unbanked applicants — the stated target population.

Subgroup accuracy: gender & age

False Positive Rate — by gender



False Positive Rate — by age group



Counter-intuitive: FPR is **lower for women** across thresholds. The model rejects creditworthy *men* more often.

Clear age bias: FPR falls steeply with age — young creditworthy applicants are rejected far more often. ROC-AUC is also lowest for the 60-70 group.

Formal fairness metrics (Fairlearn)

By gender

Threshold	DP Diff	EOpp Diff	EOdds Diff	DP Ratio
0.20	0.058	0.121	0.121	0.507
0.08	0.177	0.150	0.170	0.666

By age group

Threshold	DP Diff	EOpp Diff	EOdds Diff	DP Ratio
0.20	0.162	0.294	0.294	0.087
0.08	0.421	0.393	0.408	0.345

Age disparities » gender disparities

Every fairness metric is larger for age than for gender — by 2-3x.

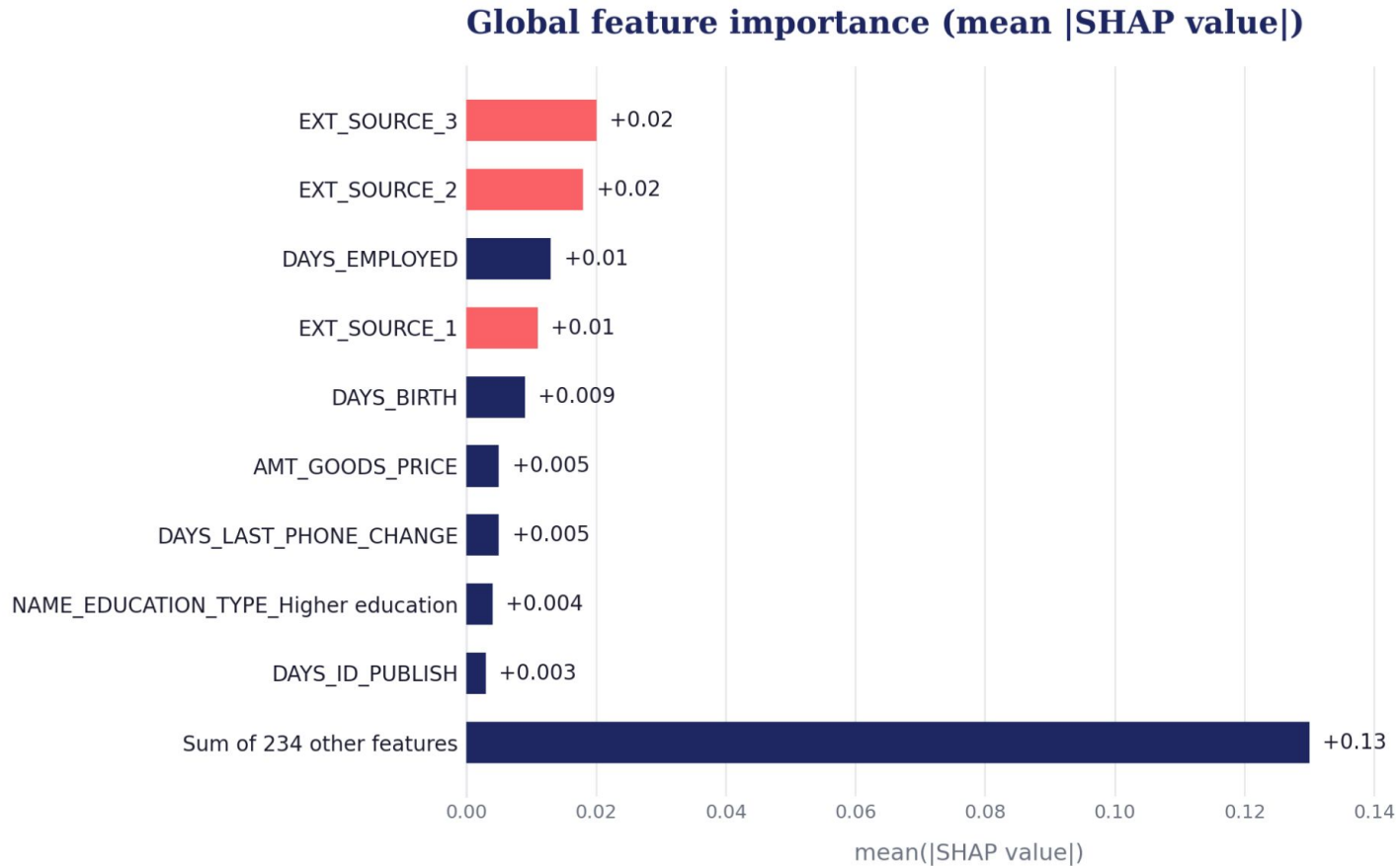
DP Ratio = 0.087

At t=0.20 the youngest age group is approved at <9% of the rate of the oldest. Far below the 0.80 "four-fifths" benchmark.

Lower threshold ⇒ larger disparities

Catching more defaulters (lower t) inflates all fairness gaps — the inclusion–risk trade-off is also a fairness trade-off.

SHAP: what drives the predictions



Takeaways

EXT_SOURCE dominates

EXT_SOURCE_2 and _3 alone contribute more than any other features combined (excluding the long tail).

Direction is consistent

Higher EXT_SOURCE → lower predicted default risk.

Inclusion concern

The model leans hardest on the signal unbanked applicants are most likely to lack.

How does the ADS treat the unbanked?

Definition. "Unbanked" = applicants missing *any* of the three EXT_SOURCE variables. Median-imputed before training.

Group	n	True default rate	Predicted default rate	ROC-AUC	FPR	FNR
Banked	21,958	0.072	0.484	0.718	0.463	0.250
Unbanked	39,545	0.085	0.471	0.702	0.446	0.271

Threshold = 0.08 used for predicted default rate, FPR, FNR.

64%

of applicants are unbanked

The audited population is the majority — not the edge case.

−0.016

ROC-AUC gap

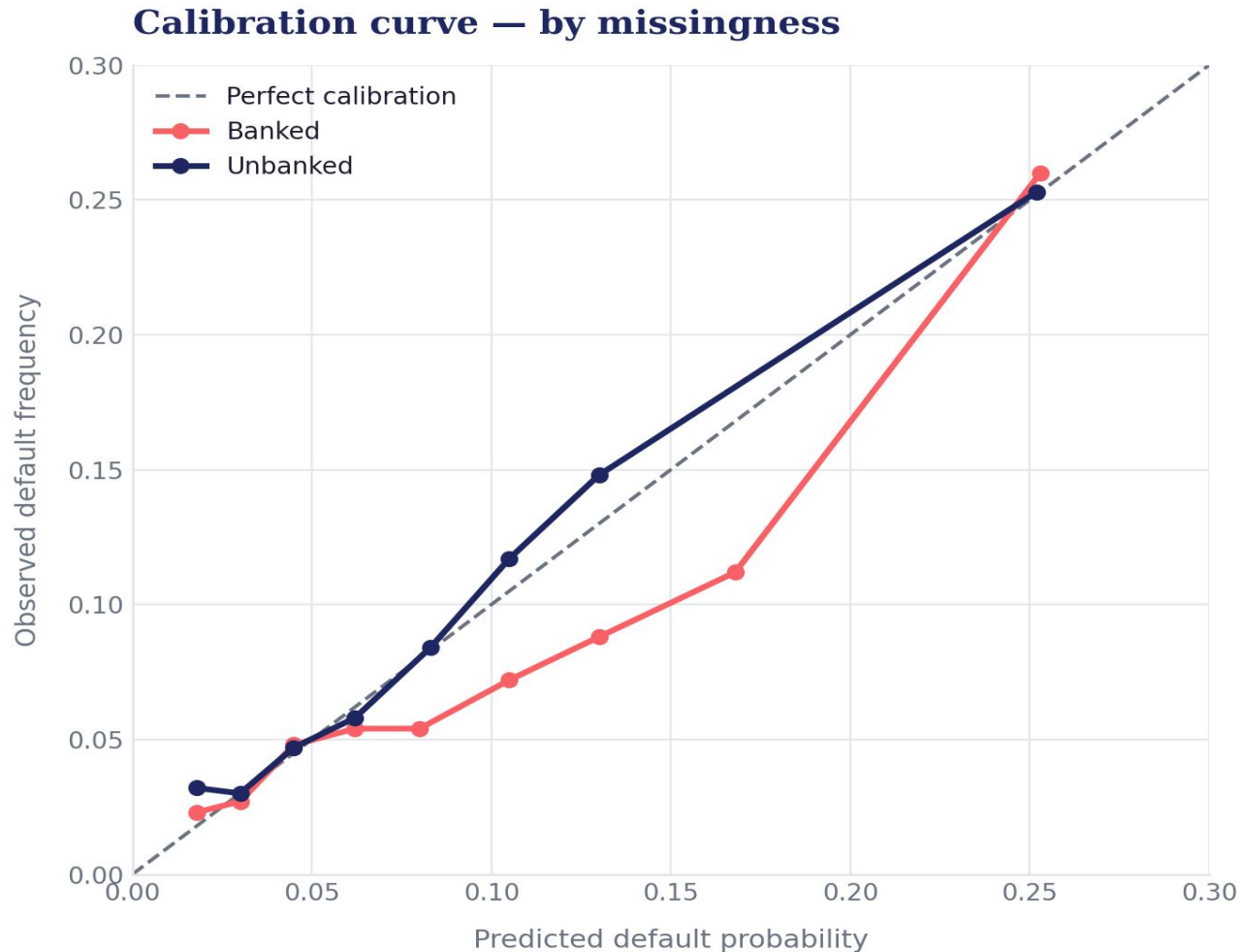
The model ranks risk noticeably worse for unbanked applicants.

+8%

higher FNR

More risky unbanked borrowers slip through undetected.

Calibration: probabilities vs. reality



What the curves show

Overall: conservative model

Both curves sit *below* the diagonal. The model over-estimates risk.

Banked group is worse-calibrated

Larger gap from the diagonal in the mid-probability range.

Unbanked closer to diagonal

But ranking is worse (lower ROC-AUC). Calibration $\neq$ discrimination.

Is the data appropriate?

Verdict: Partially appropriate. Large, real-world, and contains the demographics, financials, and alternative external scores a credit model needs — but with structural flaws.

Self-selected sample

Survivor population

Only people who applied for a Home Credit loan appear; TARGET is observed only for those who were approved. The model learns from a population shaped by past lending decisions.

Severe missingness & proxies

56% missing

67 columns contain missing values; EXT_SOURCE_1 missing for 56% of applicants. DAYS_BIRTH and CODE_GENDER correlate with TARGET — "neutral" features act as proxies for protected attributes.

Threshold choice is high-stakes

8% positive rate

The severe class imbalance makes threshold selection consequential, yet the data offers no guidance on where to set it. Lender choices propagate directly into fairness outcomes.

Robust, accurate, and fair?

Reasonably good predictive performance. Relatively stable ROC-AUC across groups; predicted probabilities broadly track observed default rates.

...but concerns about fairness and robustness remain

01

Subgroup disparities

Fairness differences across demographic groups — especially across age groups — and these gaps shift under different threshold settings.

02

Threshold trade-off

Lower thresholds reduce false negative rates but increase false positive rates and amplify fairness disparities. Inclusion–risk is also a fairness trade-off.

03

Opaque-feature reliance

Heavy reliance on the EXT_SOURCE variables, which have many missing values. Median imputation makes predictions less reliable for applicants without traditional credit history.

Metrics, stakeholders, deployment

Why these metrics?

Metric	What it captures	Primary stakeholder
ROC-AUC	Overall ranking quality	Lender
FPR	Rejecting trustworthy applicants	Applicants
FNR	Letting risky borrowers through	Lender
DP / EOpp	Disparities across groups	Applicants, regulators
Missingness audit	Reliability for unbanked	Underserved applicants

Would we deploy this ADS?

Somewhat cautious.

We would not deploy this ADS in industry or the public sector without additional review and monitoring — particularly for financially underserved populations and across different threshold settings.

Recommended improvements

→ Improve handling of missing values

Move beyond median imputation; treat missingness as informative rather than something to smooth over.

→ Reduce reliance on opaque external scores

Lessen the model's dependence on EXT_SOURCE features so unbanked applicants get more reliable predictions.

Predictively reasonable, but balancing accuracy, fairness, and inclusion remains a challenge for automated lending systems.

Thank you

Code & data sources

Home Credit Group · kaggle.com/competitions/home-credit-default-risk

Will Koehrsen · kaggle.com/code/willkoehrsen/start-here-a-gentle-introduction

Audit tools · *Fairlearn, SHAP, scikit-learn*